

R CODE LAB EXPERIMENT

SET 14 – SURVEY RESULTS

```
respondent <- 1:3
```

```
q1 <- c("A","B","C")
```

```
q2 <- c("B","A","D")
```

```
q3 <- c("C","A","B")
```

```
survey <- rbind(table(q1),table(q2),table(q3))
```

```
barplot(survey,  
        beside=FALSE,  
        col=rainbow(4),  
        legend=rownames(survey),  
        main="Survey Responses",  
        xlab="Questions",  
        ylab="Count")
```

```
install.packages("fmsb")
```

```
library(fmsb)
```

```
radar <- data.frame(  
  A=c(3,1,2),  
  B=c(3,2,1),  
  C=c(3,1,1),  
  D=c(3,0,1)  
)
```

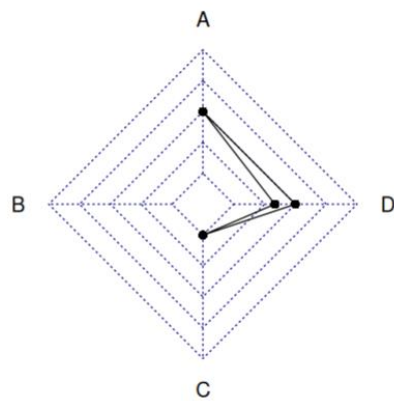
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```
radarchart(radar)
```

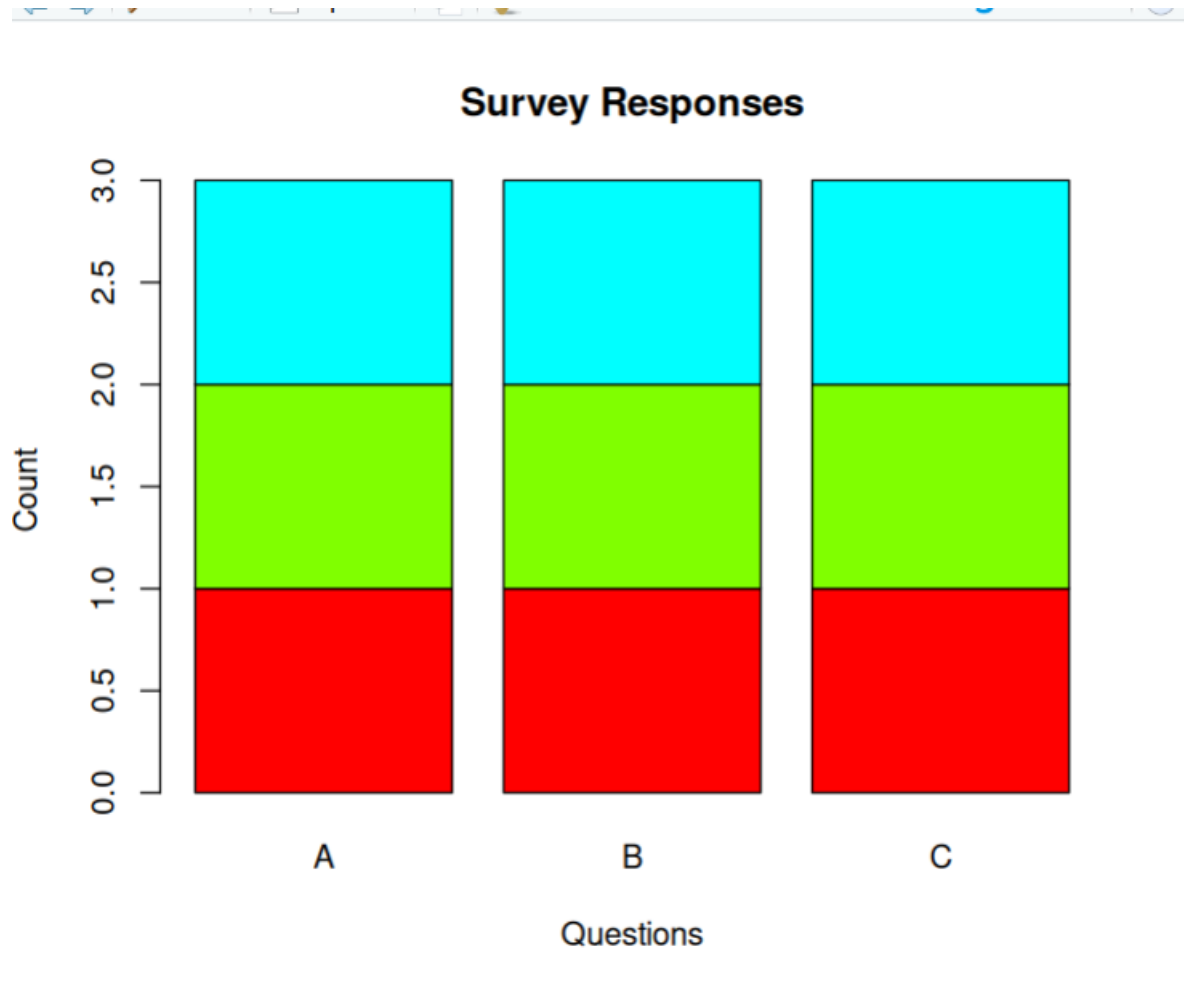
```
survey_table <- data.frame(  
  Respondent=respondent,  
  Question1=q1,  
  Question2=q2,  
  Question3=q3  
)
```

```
print(survey_table)
```

```
> print(survey_table)  
  Respondent Question1 Question2 Question3  
1          1         A         B         C  
2          2         B         A         A  
3          3         C         D         B  
> |
```



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SET 15 – PRODUCT SALES ANALYSIS

```
product <- c("Product A","Product B","Product C")
```

```
jan <- c(2000,1500,1200)
```

```
feb <- c(2200,1800,1400)
```

```
mar <- c(2400,1600,1100)
```

```
sales <- data.frame(
```

```
January=jan,
```

```
February=feb,
```

```
March=mar
```

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)

```
rownames(sales) <- product
```

```
barplot(as.matrix(sales),  
        beside=TRUE,  
        col=c("red", "green", "blue"),  
        legend=colnames(sales),  
        main="Quarterly Product Sales",  
        xlab="Products",  
        ylab="Sales")
```

```
matplot(t(sales),  
        type="l",  
        lty=1,  
        lwd=2,  
        col=1:3,  
        xaxt="n",  
        xlab="Month",  
        ylab="Sales",  
        main="Sales Trend")
```

```
axis(1,at=1:3,labels=c("Jan", "Feb", "Mar"))
```

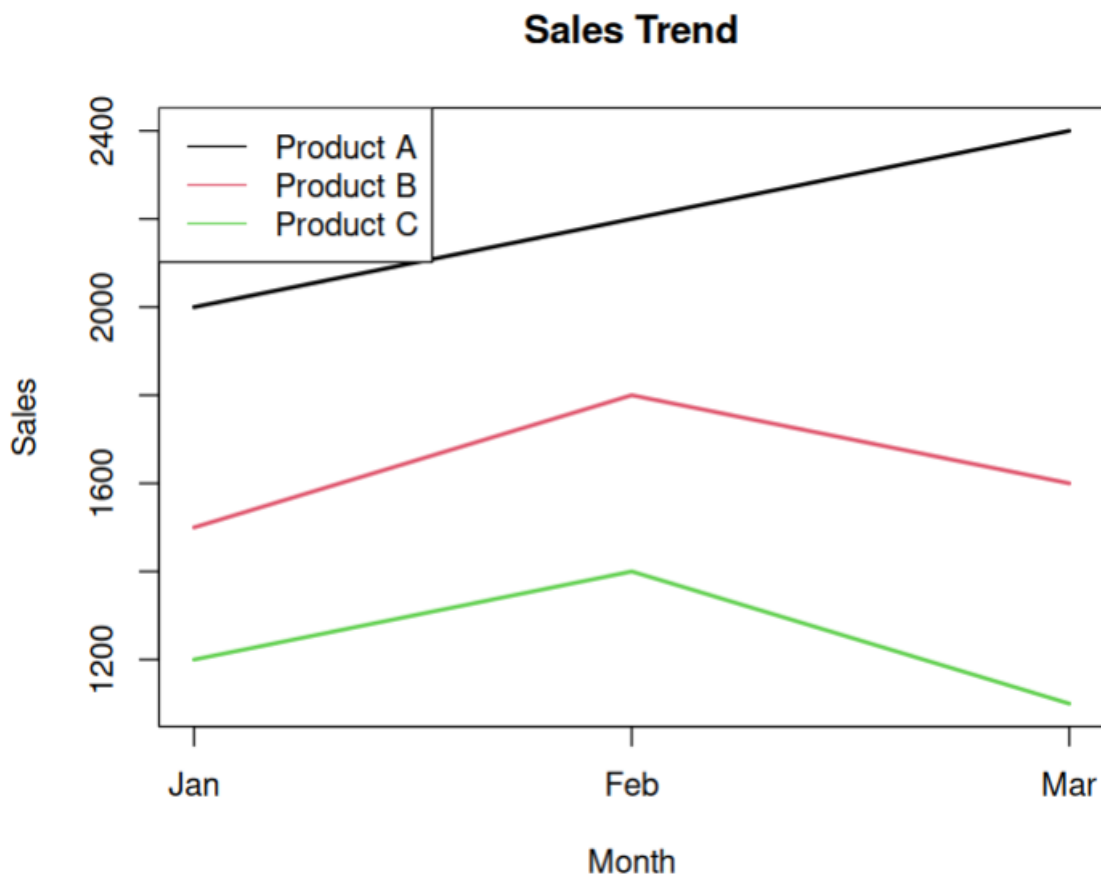
```
legend("topleft",  
       legend=product,  
       col=1:3,
```

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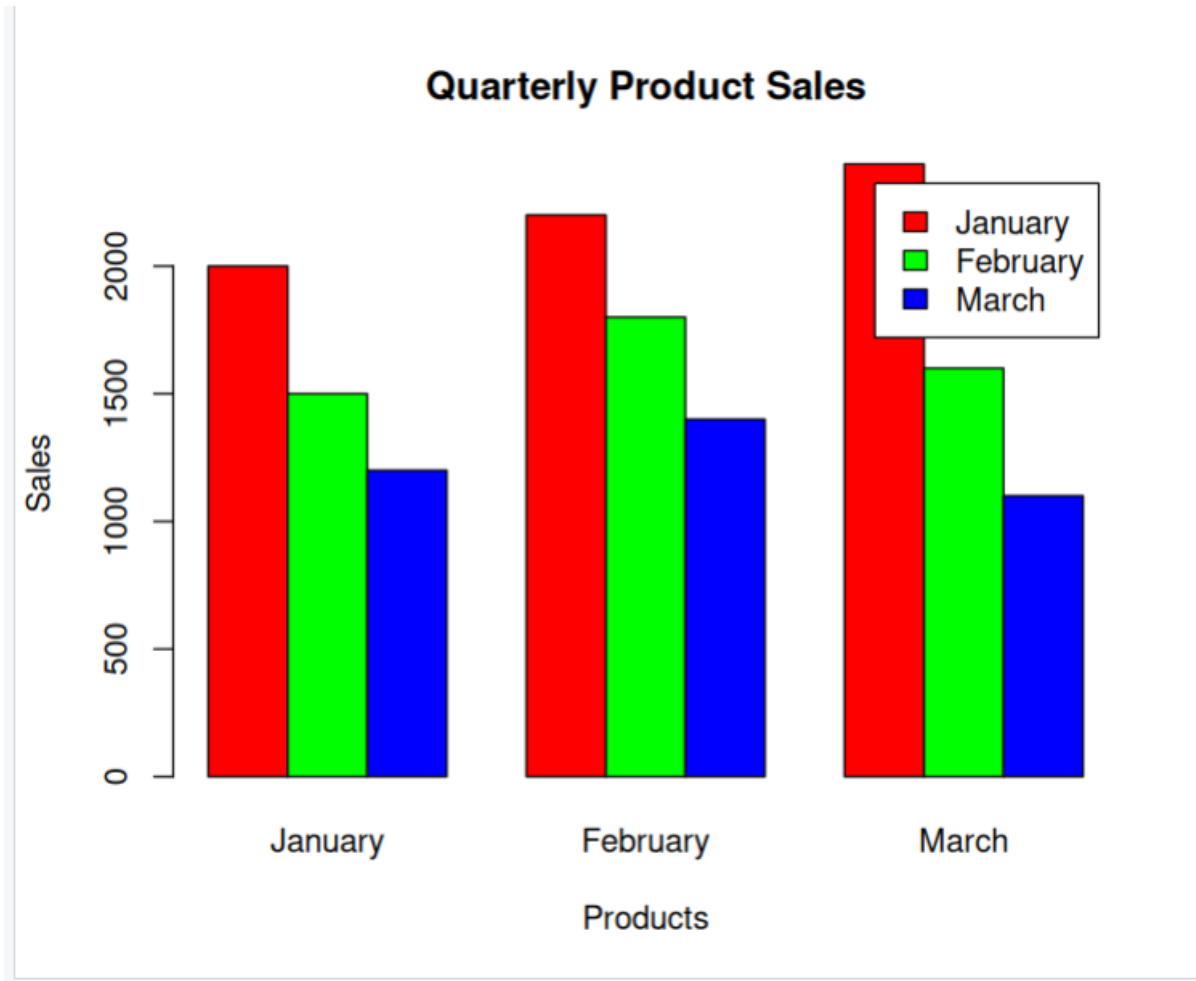
```
lty=1)
```

```
print(sales)
```

```
      January February March  
Product A    2000    2200  2400  
Product B    1500    1800  1600  
Product C    1200    1400  1100
```



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SET 16 – CUSTOMER DEMOGRAPHICS ANALYSIS

```
customer <- 1:3
```

```
age <- c(28,35,42)
```

```
gender <- c("Female","Male","Female")
```

```
income <- c(50000,60000,75000)
```

```
barplot(age,
```

```
names.arg=customer,
```

```
col="skyblue",
```

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```
main="Customer Age Distribution",
```

```
xlab="Customer ID",
```

```
ylab="Age")
```

```
pie(table(gender),
```

```
col=c("pink","lightblue"),
```

```
main="Gender Distribution")
```

```
demo <- data.frame(
```

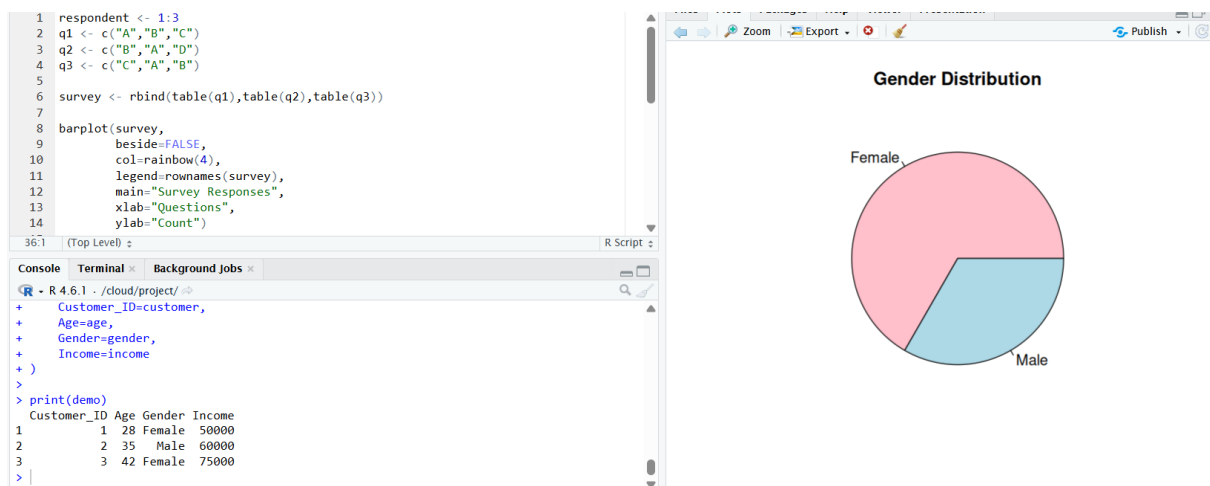
```
Customer_ID=customer,
```

```
Age=age,
```

```
Gender=gender,
```

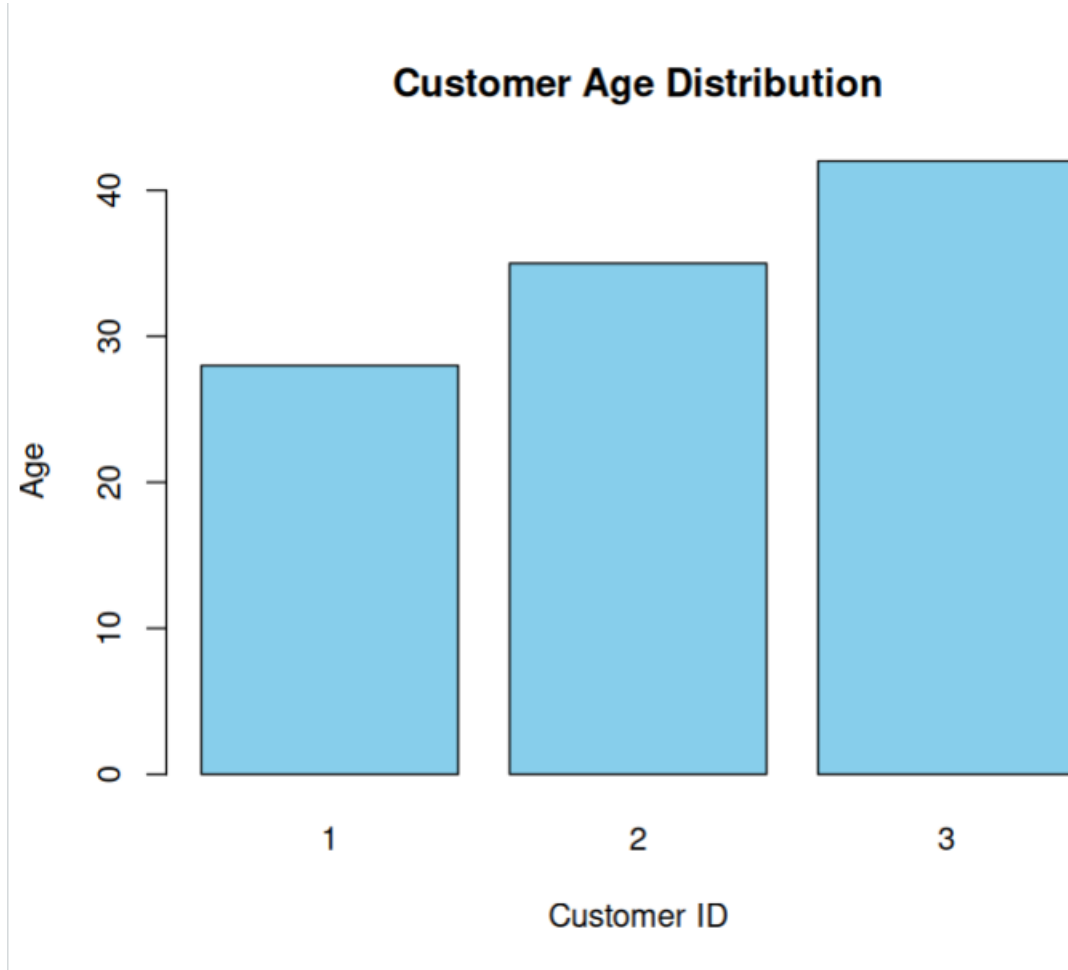
```
Income=income
```

```
)
```



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```
print(demo)
```



SET 17 – EMPLOYEE PERFORMANCE ANALYSIS

```
employee <- 1:3
```

```
department <- c("Sales","HR","Marketing")
```

```
service <- c(5,3,7)
```

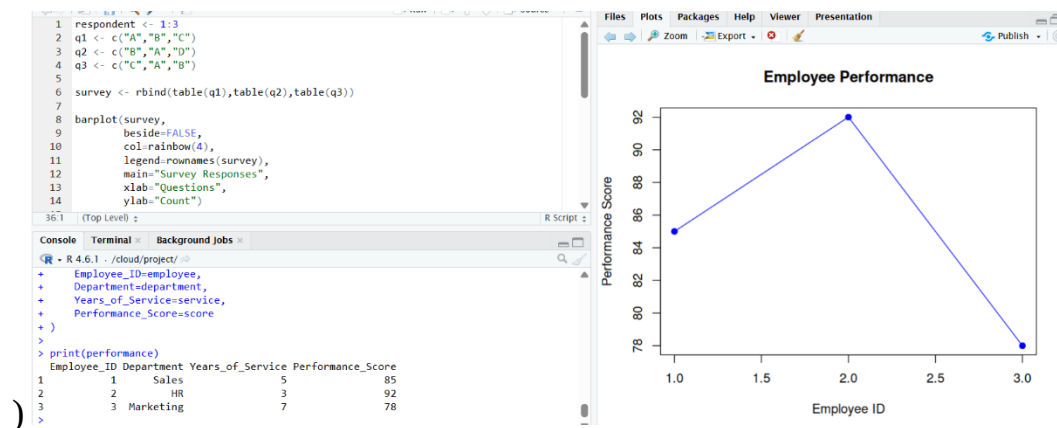
```
score <- c(85,92,78)
```

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```
plot(employee,  
score,  
type="o",  
pch=19,  
col="blue",  
main="Employee Performance",  
xlab="Employee ID",  
ylab="Performance Score")
```

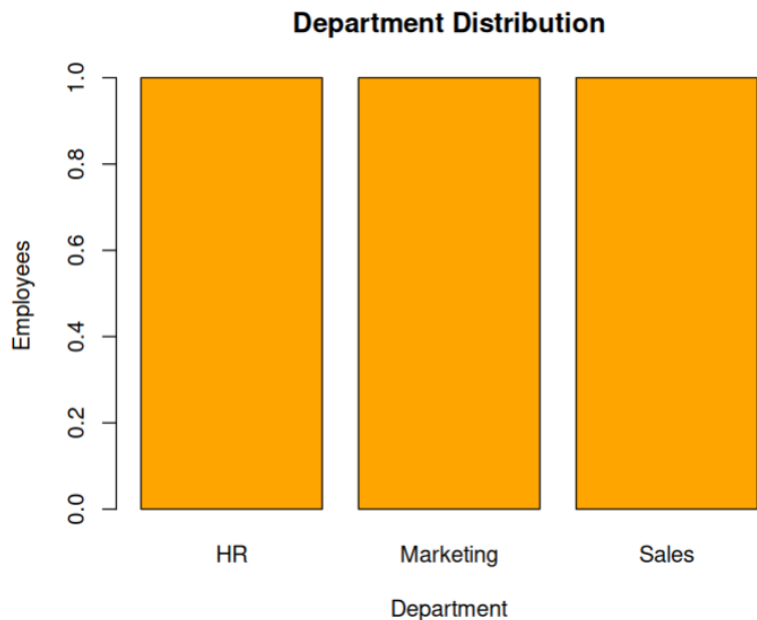
```
barplot(table(department),  
col="orange",  
main="Department Distribution",  
xlab="Department",  
ylab="Employees")
```

```
performance <- data.frame(  
Employee_ID=employee,  
Department=department,  
Years_of_Service=service,  
Performance_Score=score
```



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```
print(performance)
```



SET 18 – PRODUCT INVENTORY MANAGEMENT

```
product <- c("Product A","Product B","Product C")
```

```
quantity <- c(250,175,300)
```

```
price <- c(20,15,18)
```

```
barplot(quantity,  
names.arg=product,  
col="green",  
main="Inventory Quantity",  
xlab="Products",  
ylab="Quantity")
```

```
category1 <- c(120,90,150)
```

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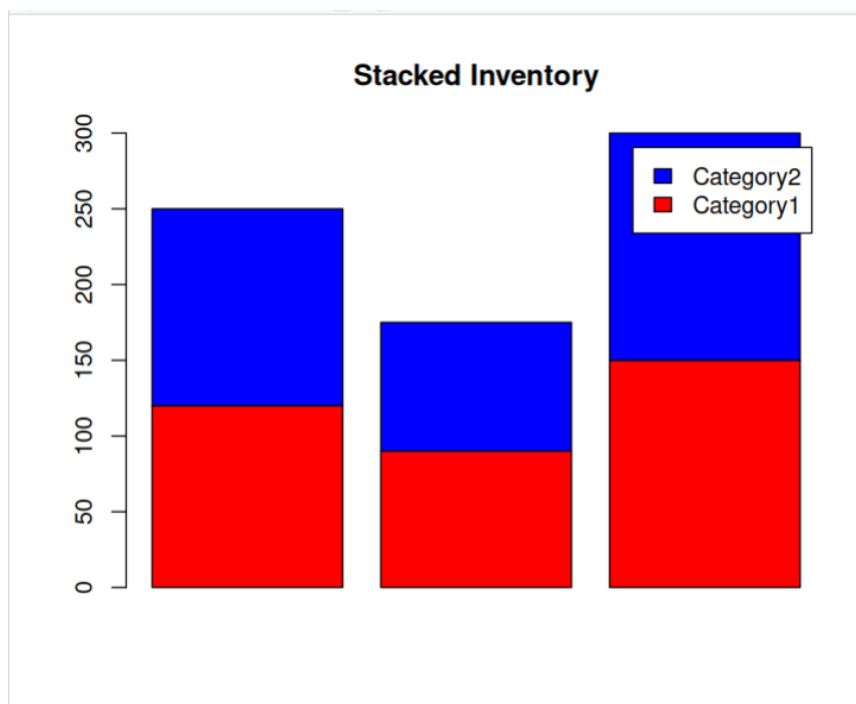
```
category2 <- quantity-category1
```

```
inventory <- rbind(category1,category2)
```

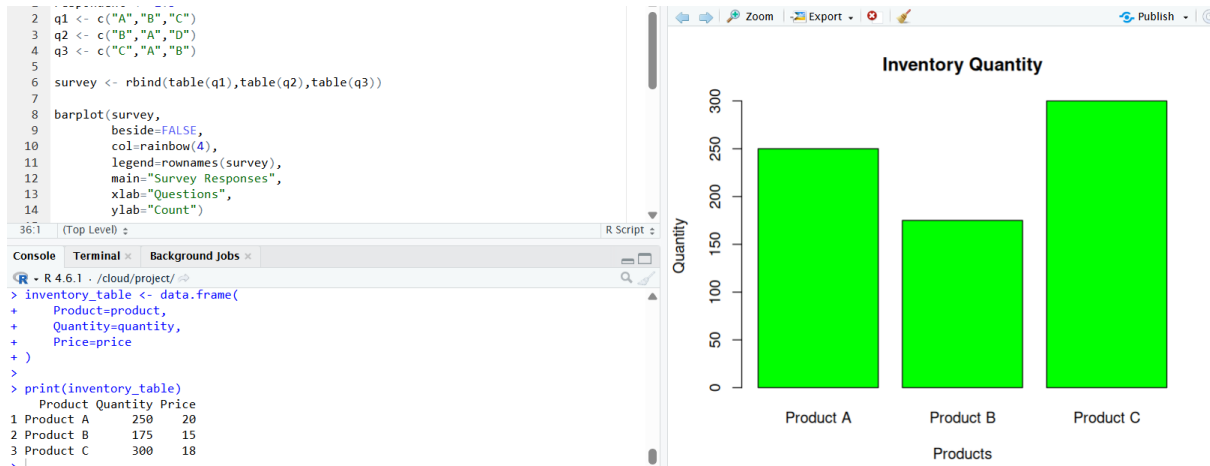
```
barplot(inventory,  
col=c("red","blue"),  
legend=c("Category1","Category2"),  
main="Stacked Inventory")
```

```
inventory_table <- data.frame(  
Product=product,  
Quantity=quantity,  
Price=price  
)
```

```
print(inventory_table)
```



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SET 19 – SURVEY RESPONSES ANALYSIS

```
survey_id <- 1:3
```

```
q1 <- c("A","B","C")
```

```
q2 <- c("B","A","D")
```

```
q3 <- c("C","A","B")
```

```
q1_table <- table(q1)
```

```
barplot(q1_table,
        beside=TRUE,
        col="skyblue",
        main="Distribution of Question 1 Responses",
        xlab="Responses",
        ylab="Count")
```

```
response_table <- rbind(table(q1),table(q2),table(q3))
```

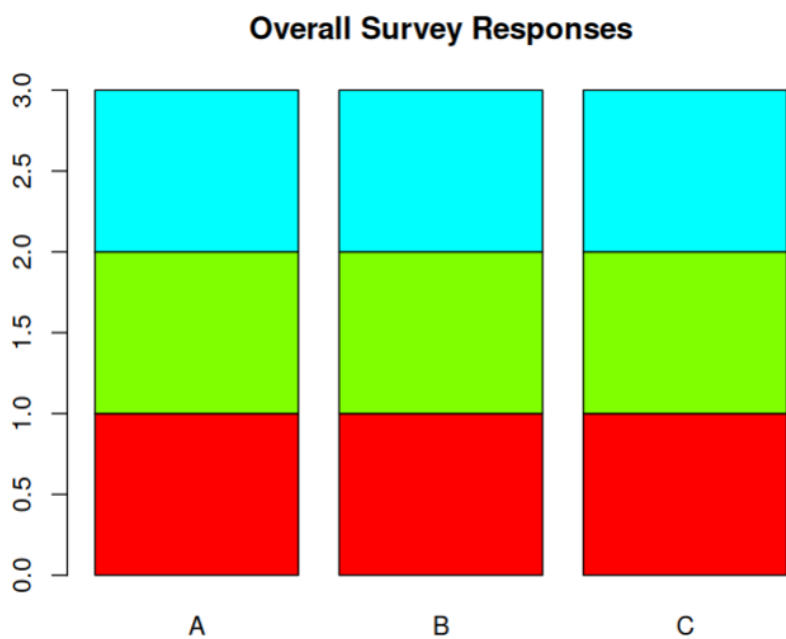
```
barplot(response_table,
```

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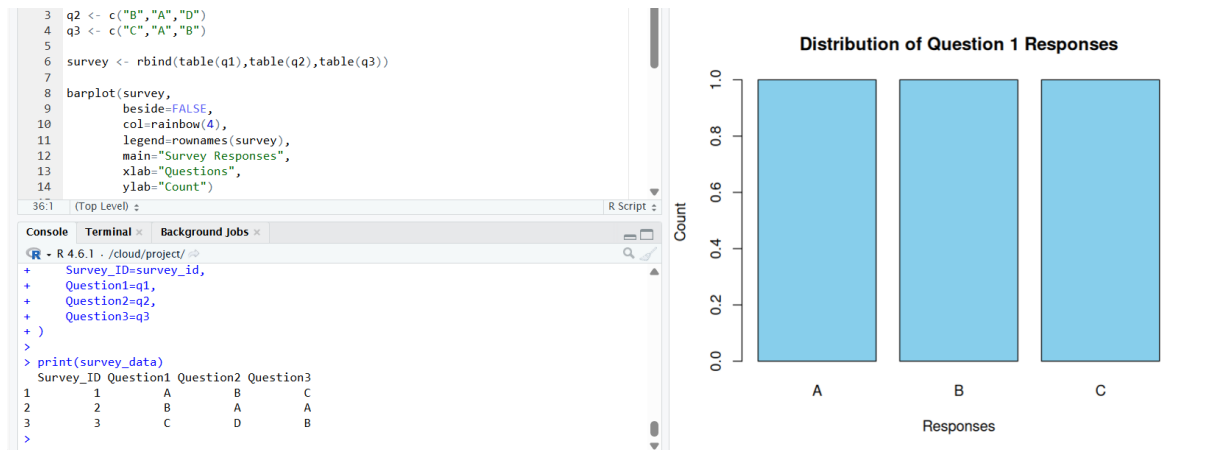
```
beside=FALSE,  
col=rainbow(4),  
legend=rownames(response_table),  
main="Overall Survey Responses")
```

```
survey_data <- data.frame(  
  Survey_ID=survey_id,  
  Question1=q1,  
  Question2=q2,  
  Question3=q3  
)
```

```
print(survey_data)
```



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SET 20 – STOCK ANALYSIS

```
date <- as.Date(c("2023-01-01","2023-01-02","2023-01-03"))
```

```
stockA <- c(100,105,110)
```

```
stockB <- c(150,152,148)
```

```
stockC <- c(120,118,122)
```

```
plot(date,
      stockA,
      type="o",
      col="red",
      ylim=c(90,160),
      xlab="Date",
      ylab="Stock Price",
      main="Stock Price Analysis")
```

```
lines(date,stockB,type="o",col="blue")
```

```
lines(date,stockC,type="o",col="green")
```

```
legend("topleft",
```

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```
legend=c("Stock A","Stock B","Stock C"),  
col=c("red","blue","green"),  
lty=1,  
pch=1)
```

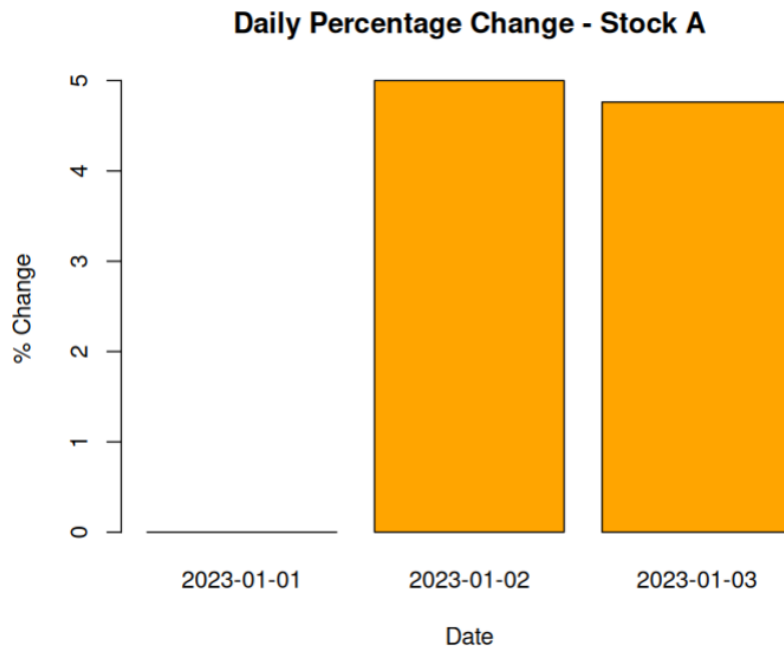
```
change <- c(0,diff(stockA)/head(stockA,-1)*100)
```

```
barplot(change,  
names.arg=date,  
col="orange",  
main="Daily Percentage Change - Stock A",  
xlab="Date",  
ylab="% Change")
```

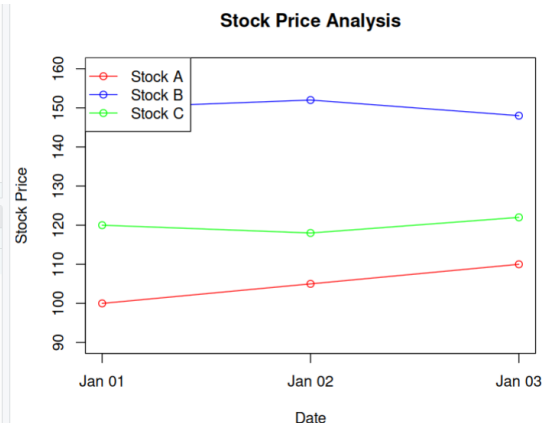
```
stock_data <- data.frame(  
Date=date,  
Stock_A=stockA,  
Stock_B=stockB,  
Stock_C=stockC  
)
```

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```
print(stock_data)
```



```
# q3 <- c("C", "A", "B")
#
# survey <- rbind(table(q1), table(q2), table(q3))
#
# barplot(survey,
#         beside=FALSE,
#         col=rainbow(4),
#         legend=rownames(survey),
#         main="Survey Responses",
#         xlab="Questions",
#         ylab="Count")
#
# (Top Level)
#
# R Script
#
# sole Terminal Background Jobs
#
# R 4.6.1 . /cloud/project/
#
# date <- as.Date(c
#   Stock_B=stockB,
#   Stock_C=stockC
# )
#
# print(stock_data)
#
# Date Stock_A Stock_B Stock_C
# 2023-01-01 100 150 120
# 2023-01-02 105 152 118
# 2023-01-03 110 148 122
```



SET 21 – ENERGY CONSUMPTION DATA

```
sector <-
c("Residential", "Commercial", "Industrial", "Residential", "Commercial", "Industrial")
```

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```
region <- c("North","South","West","East","North","South")
```

```
month <- c("Jan","Jan","Feb","Feb","Mar","Mar")
```

```
temperature <- c(15,24,20,18,28,30)
```

```
units <- c(320,540,880,350,610,920)
```

```
cost <- c(2100,3600,5900,2300,4100,6200)
```

```
renewable <- c(22,18,12,25,20,15)
```

```
peak <- c(4,6,8,5,7,9)
```

```
hist(units,
```

```
probability=TRUE,
```

```
col="skyblue",
```

```
main="Units Consumed",
```

```
xlab="Units Consumed")
```

```
lines(density(units),col="red",lwd=2)
```

```
symbols(temperature,
```

```
units,
```

```
circles=peak,
```

```
inches=0.3,
```

```
bg=rgb(0,0,1,0.5),
```

```
xlab="Temperature",
```

```
ylab="Units Consumed",
```

```
main="Temperature vs Units")
```

```
avg <- aggregate(renewable,
```

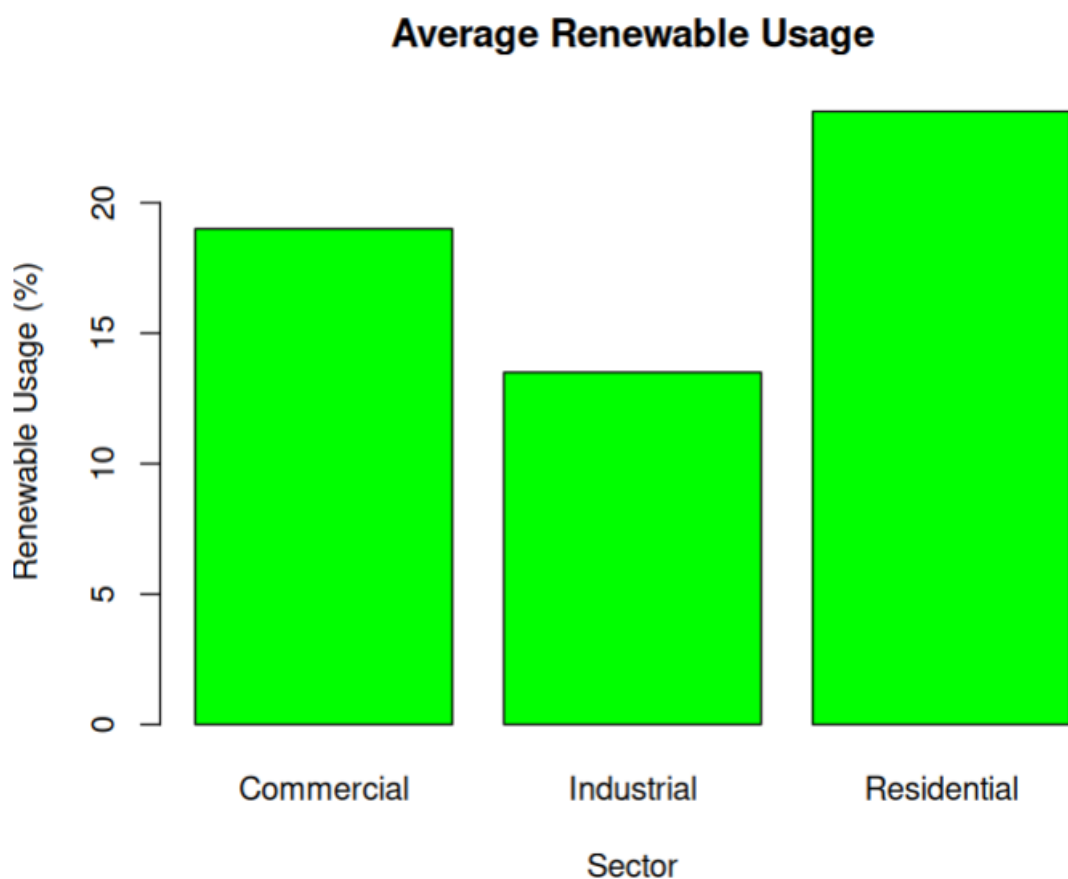
```
by=list(sector),
```

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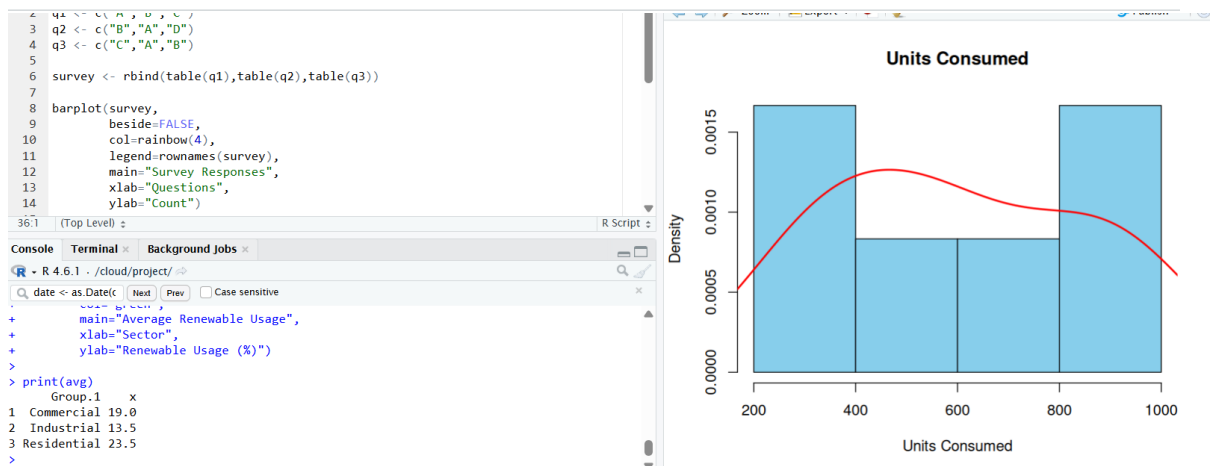
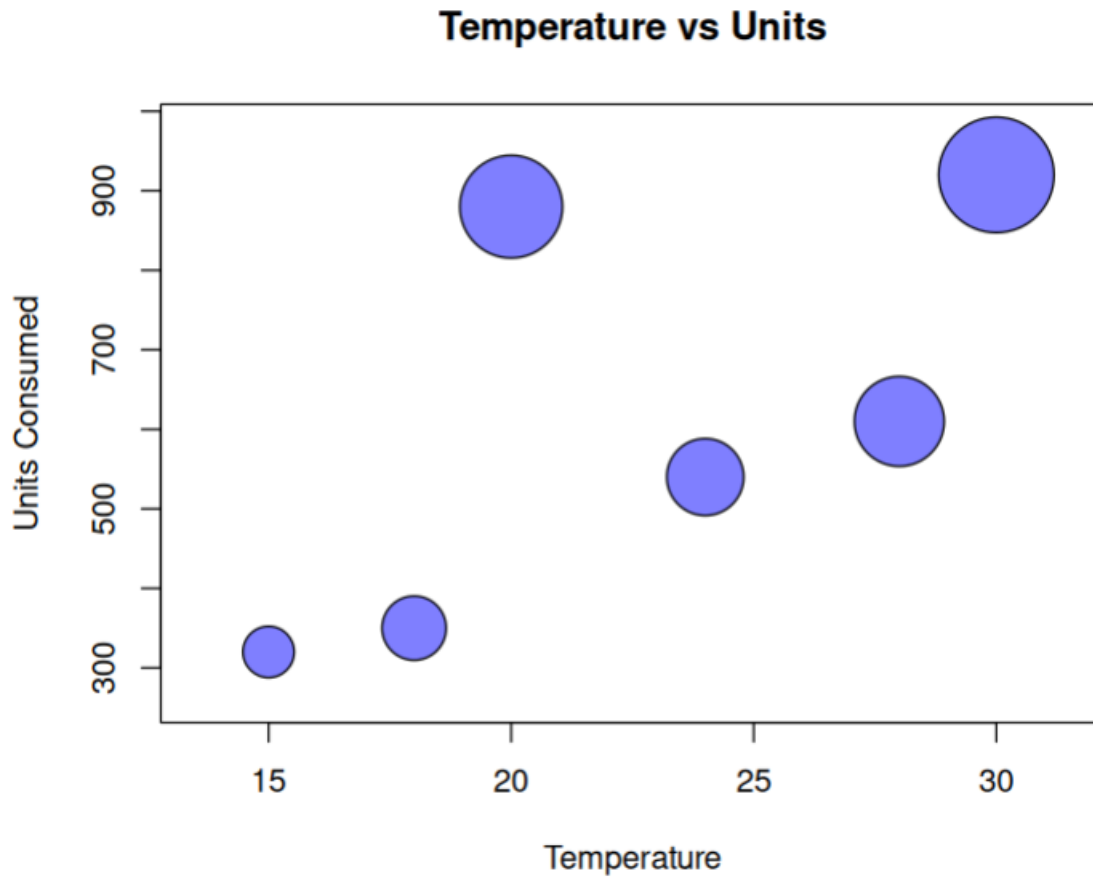
```
FUN=mean)
```

```
barplot(avg$x,  
names.arg=avg$Group.1,  
col="green",  
main="Average Renewable Usage",  
xlab="Sector",  
ylab="Renewable Usage (%)")
```

```
print(avg)
```



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SET 22 – TIME SERIES ANALYSIS

```
month <- c("January","February","March","April","May")
```

```
sales <- c(15000,18000,22000,20000,23000)
```

```
ts_sales <- ts(sales,start=1,frequency=12)
```

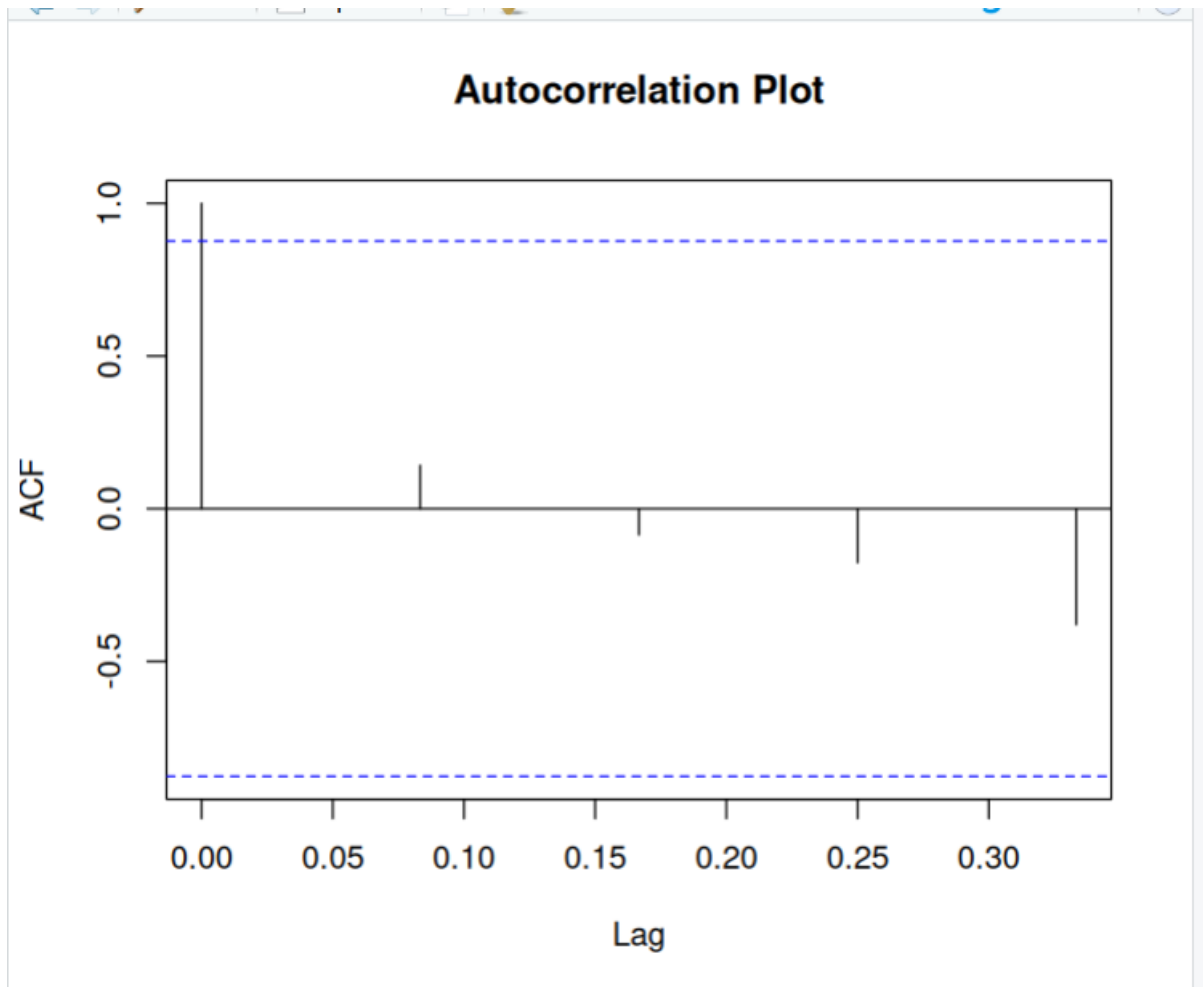
```
plot(ts_sales,  
type="o",  
col="blue",  
xlab="Month",  
ylab="Sales",  
main="Monthly Sales Trend")
```

```
advertising <- c(2000,2500,3000,2800,3200)
```

```
plot(advertising,  
sales,  
pch=19,  
col="red",  
xlab="Advertising Budget",  
ylab="Sales",  
main="Advertising Budget vs Sales")
```

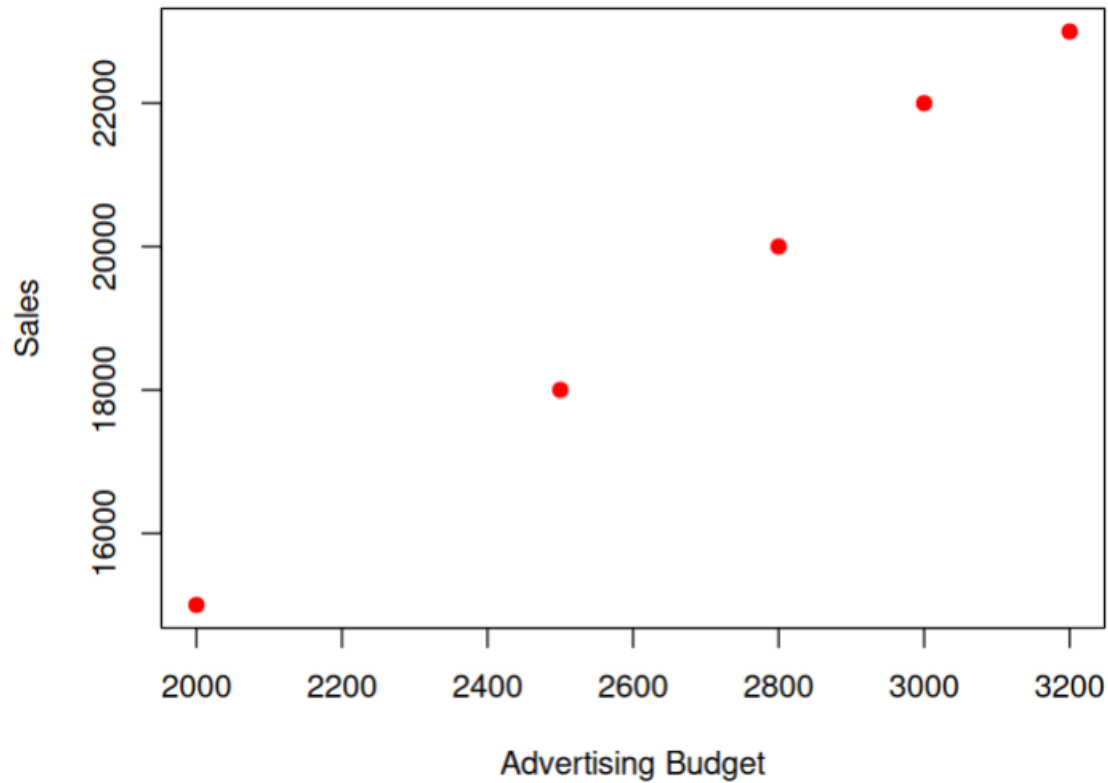
```
acf(ts_sales,  
main="Autocorrelation Plot")
```

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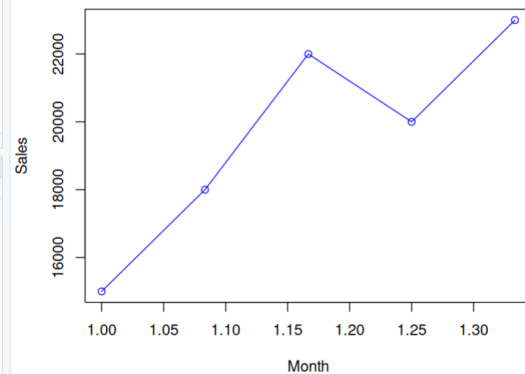
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Advertising Budget vs Sales



```
5  
6 survey <- rbind(table(q1),table(q2),table(q3))  
7  
8 barplot(survey,  
9         beside=FALSE,  
10        col=rainbow(4),  
11        legend=rownames(survey),  
12        main="Survey Responses",  
13        xlab="Questions",  
14        ylab="Count")  
15  
16 (Top Level) :  
17 R Script :  
18  
19 R 4.6.1 - /cloud/project/  
20  
21 date <- as.Date(c  
22  
23 sales,  
24 pch=19,  
25 col="red",  
26 xlab="Advertising Budget",  
27 ylab="Sales",  
28 main="Advertising Budget vs Sales")  
29  
30 acf(ts_sales,  
31     main="Autocorrelation Plot")
```

Monthly Sales Trend



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SET 23 – EMPLOYEE PERFORMANCE ANALYSIS

```
employee <- 1:5
```

```
department <- c("Sales","HR","Marketing","Sales","HR")
```

```
service <- c(5,3,7,4,2)
```

```
score <- c(85,92,78,90,76)
```

```
barplot(table(department),  
col="orange",  
main="Department Distribution",  
xlab="Department",  
ylab="Employees")
```

```
plot(employee,  
score,  
type="o",  
pch=19,  
col="blue",  
xlab="Employee ID",  
ylab="Performance Score",  
main="Employee Performance")
```

```
performance <- data.frame(  
Employee_ID=employee,  
Department=department,  
Years_of_Service=service,
```

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Performance_Score=score

)

print(performance)

